An Analysis of LLaMA language model

LLaMA language model

LLaMA language model -- Part 1

Leak On March 3, 2023, a torrent containing Llama's weights was uploaded, with a link to the torrent shared on the 4chan imageboard and subsequently spread through online AI communities. That same day, a pull request on the main Llama repository was opened, requesting to add the magnet link to the official documentation. On March 4, a pull request was opened to add links to HuggingFace repositories containing the model. On March 6, Meta filed takedown requests to remove the HuggingFace repositories linked in the pull request, characterizing it as "unauthorized distribution" of the model. HuggingFace complied with the requests. On March 20, Meta filed a DMCA takedown request for copyright infringement against a repository containing a script that downloaded Llama from a mirror, and GitHub complied the next day. Reactions to the leak varied. Some speculated that the model would be used for malicious purposes, such as more sophisticated spam. Some have celebrated the model's accessibility, as well as the fact that smaller versions of the model can be run relatively cheaply, suggesting that this will promote the flourishing of additional research developments. Multiple commentators, such as Simon Willison, compared Llama to Stable Diffusion, a text-to-image model which, unlike comparably sophisticated models which preceded it, was openly distributed, leading to a rapid proliferation of associated tools, techniques, and software.

LLaMA language model -- Part 2

Llama 3 On April 18, 2024, Meta released Llama 3 with two sizes: 8B and 70B parameters. The models have been pre-trained on approximately 15 trillion tokens of text gathered from “publicly available sources” with the instruct models fine-tuned on “publicly available instruction datasets, as well as over 10M human-annotated examples". Meta AI's testing showed in April 2024 that Llama 3 70B was beating Gemini Pro 1.5 and Claude 3 Sonnet on most benchmarks. Meta also announced plans to make Llama 3 multilingual and multimodal, better at coding and reasoning, and to increase its context window. Regarding scaling laws, Llama 3 models empirically showed that when a model is trained on data that is more than the "Chinchilla-optimal" amount, the performance continues to scale log-linearly. For example, the Chinchilla-optimal dataset for Llama 3 8B is 200 billion tokens, but performance continued to scale log-linearly to the 75-times larger dataset of 15 trillion tokens. During an interview with Dwarkesh Patel, Mark Zuckerberg said that the 8B version of Llama 3 was nearly as powerful as the largest Llama 2. Compared to previous models, Zuckerberg stated the team was surprised that the 70B model was still learning even at the end of the 15T tokens training. The decision was made to end training to focus GPU power elsewhere. Llama 3.1 was released on July 23, 2024, with three sizes: 8B, 70B, and 405B parameters.

LLaMA language model -- Part 3

Military In 2024, researchers from the People's Liberation Army Academy of Military Sciences (top military academy of China) were reported to have developed a military tool using Llama, which Meta Platforms stated was unauthorized due to Llama's license prohibiting the use of the model for military purposes. Meta granted the US government and US military contractors permission to use Llama in November 2024, but continued to prohibit military use by non-US entities.

LLaMA language model -- Part 4

Space Booz Allen Hamilton deployed Meta’s Llama 3.2 model aboard the International Space Station (ISS) National Labs as part of a project called Space Llama. The system runs on Hewlett Packard Enterprise’s Spaceborne Computer■2 and leverages Booz Allen’s A2E2 (AI for Edge Environments) platform, using NVIDIA CUDA■accelerated computing. Space Llama demonstrates how large language models can operate in disconnected, constrained environments such as space, enabling astronauts to retrieve and summarize documents using natural-language queries, even without internet connectivity.

LLaMA language model -- Part 5

Llama 1 models are only available as foundational models with self-supervised learning and without fine-tuning. Llama 2 – Chat models were derived from foundational Llama 2 models. Unlike GPT-4 which increased context length during fine-tuning, Llama 2 and Code Llama - Chat have the same context length of 4K tokens. Supervised fine-tuning used an autoregressive loss function with token loss on user prompts zeroed out. The batch size was 64. For AI alignment, human annotators wrote prompts and then compared two model outputs (a binary protocol), giving confidence levels and separate safety labels with veto power. Two separate reward models were trained from these preferences for safety and helpfulness using reinforcement learning from human feedback (RLHF). A major technical contribution is the departure from the exclusive use of proximal policy optimization (PPO) for RLHF – a new technique based on rejection sampling was used, followed by PPO. Multi-turn consistency in dialogs was targeted for improvement, to make sure that "system messages" (initial instructions, such as "speak in French" and "act like Napoleon") are respected during the dialog. This was accomplished using the new "Ghost attention" technique during training, which concatenates relevant instructions to each new user message but zeros out the loss function for tokens in the prompt (earlier parts of the dialog).

LLaMA language model -- Part 6

Llama 2 On July 18, 2023, in partnership with Microsoft, Meta announced Llama 2 (stylized as LLaMa 2), the next generation of Llama. Meta trained and released Llama 2 in three model sizes: 7, 13, and 70 billion parameters. The model architecture remains largely unchanged from that of Llama 1 models, but 40% more data was used to train the foundational models. Llama 2 includes foundation models and models fine-tuned for chat. In a further departure from the original version of Llama, all models are released with weights and may be used for many commercial use cases. Because Llama's license enforces an acceptable use policy that prohibits Llama from being used for some purposes, it is not open source. Meta's use of the term open-source to describe Llama has been disputed by the Open Source Initiative (which maintains The Open Source Definition) and others. Code Llama is a fine-tune of Llama 2 with code specific datasets. 7B, 13B, and 34B versions were released on August 24, 2023, with a 70B version released on January 29, 2024. Starting with the foundation models from Llama 2, Meta AI would train an additional 500B tokens of code datasets, before an additional 20B token of long-context data, creating the Code Llama foundation models. This foundation model was further trained on 5B instruction following token to create the instruct fine-tune. Another foundation model was created for Python code, which trained on 100B tokens of Python-only code, before the long-context data.